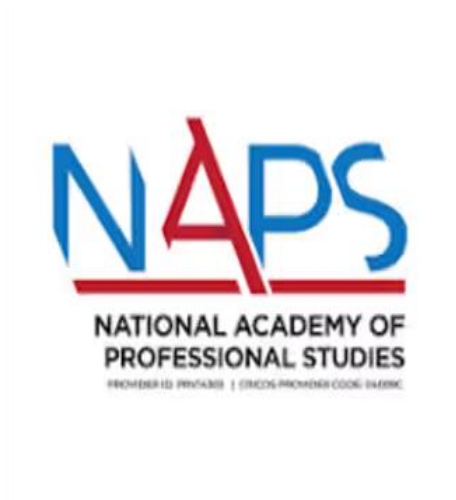


Assessment C

Milestone 1: Project Idea Proposal



Unit : Object-Oriented Design and Programming (ITS203)

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Project Title: Event Party Rental System

1.Problem Description

Event planners, both amateur and professional, have difficulty keeping track of rentals in bulk, such as chairs and tables, as well as audiovisual equipment. Planners usually solve this issue by maintaining spreadsheets, which can lead to double bookings, loss of inventory, and inability to bill clients for stolen or damaged goods. Who are the target users of your product? Small event rental business owners and managers who need to keep track of their inventory and customer's orders.

2.Proposed Idea

The proposed solution is a Windows Forms desktop application developed in C#. The application is an Event Party Rental System that allows the business owner to store information about available items for rent, maintain a database of customers, and process a rental order. It will also ensure that the same equipment is not rented twice by storing information about currently rented items in the warehouse.

3.Inspiration

I once had to create a bulk inventory management system, but I wanted to take these logical principles and apply them to a new domain. Event rentals are a great opportunity to do just that by implementing date-based availability and introducing different categories of equipment that will allow me to fully utilize C# object-oriented design patterns.

4.Functional Requirements

- The application can allow the user to add, edit, and delete inventory items (e.g., categorize them as Furniture, Linens, or Electronics).
- The application can keep track of the total quantity of an item and calculate how many are available for rent.
- The application can allow the user to make and save a new customer profile.
- The application can allow the user to make a rental booking, which links a customer to specific inventory items and dates.
- The application can calculate the total rental fee based on the item type and quantity.
- The application can process the return of items, which restores the stock of available items and alerts the user of any missing unit.

5.Non-Functional Requirements

Usability: The interface of Windows Forms should be intuitive to the end-user, who would not need to refer to a training manual to process an order.

Reliability: The application should prevent the user from entering orders for more items than are physically available, using reliable exception handling procedures.

6.Data Storage Plan

To guarantee data integrity, a relational database will be used to store the data. The tables below will be created by using standard SQL:

- **Customers Table:** CustomerID, Name, Phone, Email.

(The above table will hold information about the customers)

- **Inventory Table:** ItemID, Name, Category (Furniture, AV etc.), Total Stock, Daily Rental Rate.
(The table will contain all items available for rent and their details.)
- **Bookings Table:** BookingID, CustomerID, Rental Date, Return Date.
(This table will be used to record bookings and will record the customer who made the booking and the rental dates.)
- **The Booking_LineItems Table:** This table will hold the booking ID, item ID and quantity booked and will be used to create the many-to-many relationship between Bookings and Inventory tables.